

# Topology-Independent Real-Time AC Optimal Power Flow using Physics-Informed Fourier Neural Operators

Roshan Kumar Sharma<sup>a</sup>, Ujjwal Prasad Dahal<sup>a</sup>

<sup>a</sup>Department of Electrical Engineering, Institute of Engineering, Purwanchal Campus, Dharan, Dharan, NepalTribhuvan University

---

## Abstract

Real-time alternating-current optimal power flow (AC-OPF) is increasingly challenging in renewable-rich power systems because operating points vary rapidly and security assessment requires repeated solutions under line contingencies. Classical nonlinear programming remains computationally demanding, whereas data-driven surrogates and conventional physics-informed neural networks (PINNs) are commonly tied to a fixed bus-admittance matrix and may require re-training after a topology change. This paper proposes a topology-conditioned physics-informed graph Fourier neural operator (GFNO-PINO) that maps network topology, nodal loads, and renewable injections directly to voltage phasors and generator dispatch. Instead of applying a Euclidean fast Fourier transform to an irregular grid, the proposed operator uses Chebyshev polynomial spectral filters of the normalized graph Laplacian. Branch conductance, susceptance, status, and thermal rating are explicit inputs; consequently, an N-1 outage modifies the operator input rather than the learned parameters. Independent voltage and generator limits are enforced by bounded output reparameterizations, while AC power-balance and branch-flow equations enter a differentiable, adaptively weighted physics loss. A supervised warm start from solver labels is followed by physics-informed fine-tuning. A topology-disjoint pilot benchmark on the IEEE 30-, 57-, 118-, and 300-bus systems comprises 564 converged PowerModels-IPOPT labels and 36 held-out test scenarios per system across three unseen topologies. Relative to the data-only GFNO, physics fine-tuning reduces the mean sample-maximum KCL residual from 1.084 to 0.228, 2.467 to 1.307, 6.123 to 1.250, and 90.521 to 3.232 p.u., respectively. Median GFNO-PINO inference times are 3.66, 10.75, 4.37, and 11.31 ms, compared with 57.29, 204.45, 524.17, and 1484.22 ms for IPOPT on the same machine. However, no neural prediction satisfies the strict  $10^{-3}$  p.u. maximum-KCL feasibility proxy. Therefore, signed negative cost gaps are not interpreted as feasible optimality improvements. The spectral model also does not consistently outperform the spatial GNN across systems. These results establish useful topology-conditioned residual reduction and latency, while showing that feasibility restoration and larger multi-seed experiments remain necessary before operational deployment.

**Keywords:** FNO, AC-OPF, Physics-Informed Neural Operator, N-1 Contingency, Graph Neural Networks, Renewable Integration

---

## 1. Introduction

Power-system operation is being reshaped by variable renewable generation, electrification, and increasingly active demand. Solar and wind injections can change substantially within an operating interval, while load uncertainty alters both active- and reactive-power requirements. At the same time, the N-1 security criterion requires operators to assess whether the system remains within voltage, generation, and branch-flow limits after the loss of any credible component. These interacting sources of variability turn dispatch into a sequence of closely spaced, nonlinear, topology-dependent optimization problems rather than a single deterministic calculation. Reviews of optimal power flow (OPF) and security-constrained OPF emphasize both their operational importance and their continuing computational difficulty [1–4].

The alternating-current OPF (AC-OPF) formulation represents voltage magnitudes, phase angles, active and re-

active generation, losses, transformer effects, and thermal constraints. This fidelity produces non-convex equality and inequality constraints. Mature nonlinear solvers such as IPOPT apply sparse primal-dual interior-point methods and are indispensable for offline planning and high-quality reference solutions [5]. Nevertheless, a complete solve is still required for each loading condition and each network topology. Under large scenario ensembles or online contingency screening, solver initialization, factorization, and convergence variability can become a latency bottleneck. Convex relaxations can provide valuable bounds but introduce their own tightness and scalability tradeoffs [4].

The direct-current OPF (DC-OPF) approximation is faster because it neglects reactive power, assumes voltage magnitudes near one per unit, and linearizes angle differences. These assumptions make DC-OPF useful for screening and market studies, but they can produce material errors under high loading, resistive networks, voltage-

sensitive conditions, or binding reactive-power limits [6]. A real-time surrogate for AC-OPF should therefore be benchmarked against DC-OPF for speed without treating the linear approximation as an AC-feasible ground truth.

Learning-based OPF methods amortize the cost of optimization: an expensive solver generates training examples offline, after which a neural network provides rapid on-line predictions. DeepOPF and related methods demonstrate the potential of predict-and-reconstruct strategies, feasibility-aware training, and topology embeddings [7–9]. Other work combines supervised learning with Lagrangian penalties or worst-case verification [10, 11]. However, a finite-dimensional network trained on one fixed ordering can learn spurious correlations specific to that network. Even when admittance is an input, dense architectures do not directly encode the locality, permutation structure, or graph spectrum of an electrical network.

Physics-informed neural networks (PINNs) reduce reliance on labels by penalizing governing-equation residuals [12, 13]. In power systems, PINNs have been used to embed power-flow or optimality equations [14–16]. A standard PINN nevertheless represents a solution for a fixed physical instance or uses a fixed-dimensional parameterization. If its residual is formed with one admittance matrix, replacing a line changes the governing operator itself. Retraining a separate PINN for each contingency defeats the intended real-time advantage.

Neural operators provide a more suitable abstraction: they learn maps between function spaces rather than a single input-output vector relationship. DeepONet learns an operator through branch and trunk networks [17], while the Fourier neural operator (FNO) parameterizes non-local integral kernels in a spectral domain [18, 19]. The original FNO assumes a regular Euclidean grid and a shared Fourier basis, whereas transmission networks are sparse, irregular graphs. Graph spectral methods replace Euclidean harmonics with functions of the graph Laplacian [20–22]. Chebyshev polynomial filters are particularly attractive under contingencies: they apply the same learned spectral multiplier to a changing Laplacian without requiring a topology-specific eigendecomposition or eigenvector alignment.

This paper develops a topology-conditioned physics-informed graph Fourier neural operator (GFNO-PINO) for real-time AC-OPF. Its principal contributions are:

- A graph neural-operator formulation in which line admittance, status, and rating explicitly condition the learned mapping, allowing an N-1 outage to be processed without topology-specific retraining.
- A Chebyshev spectral GFNO architecture that preserves the spectral operator bias of FNO while avoiding unstable eigenvector coordinates across related network topologies.
- A differentiable AC physics objective using exact bus-admittance and branch-current operators, cou-

pled with adaptive gradient-norm loss balancing and a supervised-to-physics curriculum.

- Hard output transformations for voltage and generator box limits and the slack-angle reference, accompanied by explicit reporting of the AC equality and thermal constraints that remain only approximately satisfied.
- A reproducible topology-disjoint benchmark on four IEEE systems against IPOPT AC-OPF, DC-OPF, a fixed-topology PINN, a spatial GNN, and a data-only graph FNO.

The remainder of the paper is organized as follows. Section 2 reviews classical, learning-based, physics-informed, and neural-operator approaches. Section 3 presents the AC-OPF formulation and the proposed GFNO-PINO. Section 4 defines the benchmark protocol. Section 5 provides result tables and discussion scaffolds to be completed from released experiment artifacts. Section 6 states the scope of the feasibility and generalization claims, and Section 7 concludes the paper.

## 2. Related work

### 2.1. Classical and learning-based AC-OPF

AC-OPF is a foundational nonlinear optimization problem in power-system planning and operation. Standard formulations minimize generation cost subject to nonlinear nodal balances and equipment constraints [1, 2]. Sparse interior-point algorithms provide reliable local solutions and form the basis of widely used tools [5]. MATPOWER, pandapower, and PowerModels provide reproducible implementations and test-case interfaces [23–25]. Because AC-OPF is non-convex, a locally converged point is not in general a certificate of global optimality. Convex and moment relaxations may produce bounds or exact solutions in favorable cases, but no single relaxation dominates across all large practical networks [3, 4].

Supervised surrogates learn the mapping from uncertain demand to optimal decisions. DeepOPF predicts independent variables and reconstructs dependent quantities to improve feasibility [7, 8]. Fioretto et al. combine deep learning with a Lagrangian-dual objective [10]; Venzke et al. formulate optimization problems that bound worst-case neural-network violations [11]. These approaches show that inference can be much faster than solving each instance from scratch, but feasibility treatment and generalization domains differ. DeepOPF-FT is especially relevant because it embeds topology and admittance to address multiple related AC-OPF problems [9]. The present work differs by using a graph-spectral operator whose propagation rule itself changes with the input topology and by training with exact AC residuals.

## 2.2. Physics-informed neural networks in power systems

PINNs augment data losses with residuals of governing equations [12]. The broader physics-informed machine-learning framework can improve data efficiency and physical consistency, but optimization can be stiff when residual terms have different units or gradient scales [13]. In power applications, Misyris et al. embedded power-system equations in neural training [14], and Lei et al. developed a physics-informed data-driven OPF framework [16]. Nellikkath and Chatzivasileiadis incorporated AC-OPF physics and methods for reducing worst-case violations [15].

Physics penalties do not automatically provide feasibility. A small mean residual can coexist with a large worst-bus error, and weighted penalties may trade economic performance against constraint satisfaction. Furthermore, a PINN residual evaluated using one fixed  $Y_{\text{bus}}$  does not describe the equations after a branch outage. The network may be fine-tuned for every new  $Y_{\text{bus}}$ , but such a workflow is not topology independent. The proposed method instead treats topology as part of the input function and evaluates every residual with the corresponding sample-specific admittance operators.

## 2.3. Neural operators and graph spectral methods

DeepONet establishes a practical architecture for learning nonlinear operators using separate encoders for input functions and output coordinates [17]. FNO learns a global convolution kernel in Fourier space and has demonstrated mesh-transfer properties for parametric partial differential equations [18]. Neural-operator theory formalizes discretization-aware mappings between infinite-dimensional spaces [19]. Applying a Euclidean FFT directly to buses, however, would assign physical meaning to an arbitrary bus ordering and would not respect irregular connectivity.

Graph signal processing defines spectral filters through the eigenstructure of the graph Laplacian [20]. Chebyshev networks approximate such filters by stable polynomials, avoiding explicit eigendecomposition [21]. Graph convolution and geometric deep learning extend local and spectral operations to non-Euclidean domains [22, 26]. In power systems, graph neural solvers exploit network structure for power flow and OPF [27, 28]. A spatial GNN aggregates nearby features, whereas a polynomial spectral filter explicitly parameterizes a frequency response of the topology-conditioned Laplacian.

## 2.4. Research gap

The preceding literatures leave a specific gap. Classical methods account for topology and AC physics but solve every scenario separately. Supervised OPF surrogates provide low latency but can be tied to a fixed dimension and may require post-processing. PINNs inject physics but are commonly trained for a fixed admittance matrix. Neural operators target families of equations, yet

standard FNOs assume a regular grid, and existing OPF GNNs do not necessarily combine a graph-spectral operator, sample-specific AC residuals, adaptive multi-objective training, and topology-disjoint evaluation. This work addresses that intersection. “Topology independent” is used here in the operational sense of one parameter set processing the base grid and unseen non-islanding line-status configurations within a compatible network family; it is not a claim of universal zero-shot transfer to arbitrary utilities or unbounded network sizes.

## 3. Methodology

### 3.1. AC-OPF problem formulation

Let  $\mathcal{N}$ ,  $\mathcal{E}$ , and  $\mathcal{G}$  denote the bus, directed branch, and dispatchable-generator sets. Generator-to-bus incidence is represented by  $\mathcal{G}_i \subseteq \mathcal{G}$ . For bus  $i$ , the voltage phasor is  $\mathcal{V}_i = V_i e^{j\theta_i}$ . Active and reactive demand are  $P_i^d, Q_i^d$ , renewable active injection is  $P_i^r$ , and dispatchable generation is  $P_g^g, Q_g^g$ . With quadratic cost coefficients  $c_{2g}, c_{1g}, c_{0g}$ , AC-OPF is

$$\underset{V, \theta, P_g^g, Q_g^g}{\text{minimize}} \quad C(P^g) = \sum_{g \in \mathcal{G}} (c_{2g}(P_g^g)^2 + c_{1g}P_g^g + c_{0g}). \quad (1)$$

For topology  $z$ , let  $Y(z) = G(z) + jB(z) \in \mathbb{C}^{|\mathcal{N}| \times |\mathcal{N}|}$  be the bus-admittance matrix. A base topology has  $z_\ell = 1$  for every in-service line. An N-1 topology sets exactly one eligible line status to zero:

$$\mathcal{T}_{N-1} = \left\{ z \in \{0, 1\}^{|\mathcal{E}|} \left| \sum_{\ell \in \mathcal{E}} (1 - z_\ell) = 1, (\mathcal{N}, \mathcal{E}(z)) \text{ connected} \right. \right\}. \quad (2)$$

The active- and reactive-power injections implied by the voltage are

$$P_i(V, \theta; z) = V_i \sum_{j \in \mathcal{N}} V_j \left[ G_{ij}(z) \cos(\theta_i - \theta_j) + B_{ij}(z) \sin(\theta_i - \theta_j) \right], \quad (3)$$

$$Q_i(V, \theta; z) = V_i \sum_{j \in \mathcal{N}} V_j \left[ G_{ij}(z) \sin(\theta_i - \theta_j) - B_{ij}(z) \cos(\theta_i - \theta_j) \right]. \quad (4)$$

Kirchhoff’s current law, written as nodal complex-power balance, requires

$$\sum_{g \in \mathcal{G}_i} P_g^g + P_i^r - P_i^d = P_i(V, \theta; z), \quad \forall i \in \mathcal{N}, \quad (5)$$

$$\sum_{g \in \mathcal{G}_i} Q_g^g - Q_i^d = Q_i(V, \theta; z), \quad \forall i \in \mathcal{N}. \quad (6)$$

Equations (3)–(6) are implemented in complex form as

$$S^{\text{net}} = \mathcal{V} \odot (Y(z) \mathcal{V})^*, \quad (7)$$

which is algebraically equivalent and avoids materializing a dense tensor of all angle differences.

Let  $Y_f(z)$  and  $Y_t(z)$  map bus voltages to branch currents at the from and to terminals. This representation retains line charging, off-nominal transformer ratios, and phase shifts. Terminal powers are

$$S_\ell^f = \mathcal{V}_{f(\ell)} ([Y_f(z)\mathcal{V}]_\ell)^*, \quad (8)$$

$$S_\ell^t = \mathcal{V}_{t(\ell)} ([Y_t(z)\mathcal{V}]_\ell)^*. \quad (9)$$

Because all branch drops derive from a common set of bus phasors, cycle-voltage KVL is structural. For every oriented cycle  $C \subseteq \mathcal{E}$ ,

$$\sum_{(i,j) \in C} (\mathcal{V}_i - \mathcal{V}_j) = 0. \quad (10)$$

This diagnostic tests numerical consistency of the common phasor representation; it does not replace the nonlinear KCL conditions. The remaining operating constraints are

$$P_g^{\min} \leq P_g^s \leq P_g^{\max}, \quad Q_g^{\min} \leq Q_g^s \leq Q_g^{\max}, \quad \forall g \in \mathcal{G}, \quad (11)$$

$$V_i^{\min} \leq V_i \leq V_i^{\max}, \quad \forall i \in \mathcal{N}, \quad (12)$$

$$\max\{|S_\ell^f|, |S_\ell^t|\} \leq S_\ell^{\max}, \quad \forall \ell \in \mathcal{E}(z), \quad (13)$$

$$\theta_{\text{iref}} = 0. \quad (14)$$

A zero MATPOWER branch rating denotes an unconstrained line and is excluded from the thermal-violation denominator rather than interpreted as a zero-MVA limit.

### 3.2. Operator-learning formulation

For each bus, the input field is

$$x_i = [P_i^d, Q_i^d, P_i^r, \text{onehot}(\text{type}_i), V_i^{\text{set}}] \in \mathbb{R}^8. \quad (15)$$

The setpoint channel is zero at buses without a regulated generator. For each canonical branch  $\ell = (i, j)$ , the edge field is

$$e_\ell = [G_\ell, B_\ell, z_\ell, S_\ell^{\max}] \in \mathbb{R}^4, \quad (16)$$

where  $G_\ell + jB_\ell$  is the effective transfer admittance including the off-nominal tap and phase shift in the from-to direction. Exact bidirectional physics is retained in  $Y_f$  and  $Y_t$  for the loss.

The target solution field is  $u = (V, \theta, P^s, Q^s)$ . The learned operator is

$$\mathcal{G}_\Theta : (X, E, z, \mathcal{M}) \mapsto (\widehat{V}, \widehat{\theta}, \widehat{P}^s, \widehat{Q}^s), \quad (17)$$

where  $\mathcal{M}$  contains bus and generator limits, incidence maps, and the slack-bus indicator. Parameters  $\Theta$  are shared across every topology presented to a checkpoint. The graph dimensions enter through tensor axes and gather/scatter operations, rather than learned bus-specific parameters.

### 3.3. Graph Fourier neural operator architecture

For topology  $z$ , an undirected weighted adjacency matrix is assembled using

$$A_{ij}(z) = \sum_{\ell: \{f(\ell), t(\ell)\} = \{i, j\}} z_\ell \sqrt{G_\ell^2 + B_\ell^2}. \quad (18)$$

With  $D = \text{diag}(A\mathbf{1})$ , the symmetric normalized Laplacian is

$$L(z) = I - D^{-1/2} A(z) D^{-1/2}. \quad (19)$$

Its spectrum lies in  $[0, 2]$  for a nonnegative graph, so  $\widetilde{L} = L - I$  maps the spectrum to  $[-1, 1]$  without computing a sample-specific largest eigenvalue.

A direct graph-Fourier layer would multiply in the eigenbasis  $L = U\Lambda U^\top$ . Under a line outage, eigenvectors can change sign, rotate within repeated eigenspaces, or exchange order. Consequently, coefficient channels do not have a consistent interpretation across topologies. We instead parameterize the spectral response by a  $K$ -term Chebyshev polynomial:

$$\begin{aligned} T_0(\widetilde{L})H &= H, & T_1(\widetilde{L})H &= \widetilde{L}H, \\ T_k(\widetilde{L})H &= 2\widetilde{L}T_{k-1}(\widetilde{L})H - T_{k-2}(\widetilde{L})H, & k &\geq 2, \end{aligned} \quad (20)$$

$$\mathcal{K}_\Theta^{(r)}(H; z) = \sum_{k=0}^{K-1} T_k(\widetilde{L}(z)) H W_k^{(r)}. \quad (21)$$

This is a graph-spectral multiplier, but it is basis-free and localized to  $K-1$  hops. It requires no eigendecomposition for a new contingency.

An edge multilayer perceptron first encodes  $e_\ell$ ; its output is multiplied by  $z_\ell$  so that the bias cannot leak information through an outaged line. The encoded edges are averaged at their incident buses and concatenated with  $x_i$ . A linear lifting map produces width- $d$  features  $H^{(0)}$ . Each GFNO block combines the polynomial spectral path with a learned pointwise path:

$$\widetilde{H}^{(r+1)} = \text{GELU}(\mathcal{K}_\Theta^{(r)}(H^{(r)}; z) + H^{(r)} W_{\text{pw}}^{(r)} + b^{(r)}), \quad (22)$$

$$H^{(r+1)} = \text{LayerNorm}(H^{(r)} + \text{Dropout}(\widetilde{H}^{(r+1)})). \quad (23)$$

Bus heads decode voltage magnitude and angle. Generator heads gather the latent feature at each generator bus, concatenate generator-specific limits and voltage setpoint, and decode active and reactive dispatch.

### 3.4. Physics-informed loss function

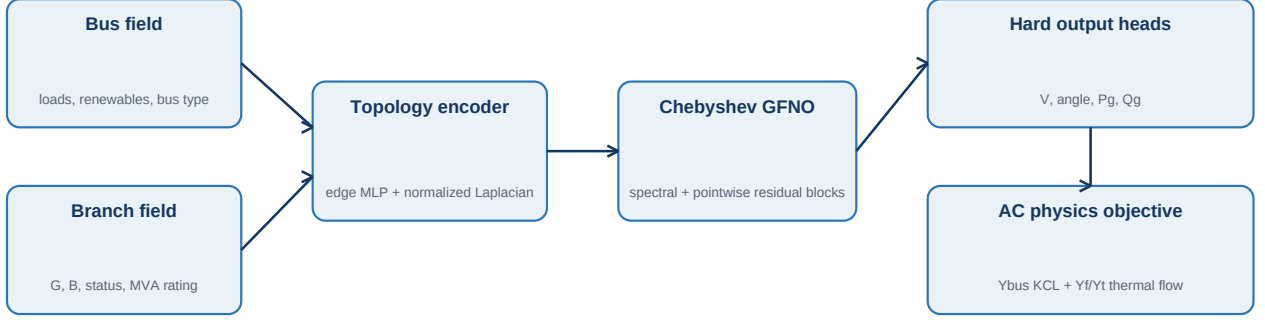
For a batch  $\mathcal{B}$ , the economic term is

$$\mathcal{L}_{\text{economic}} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} C(\widehat{P}_b^s). \quad (24)$$

Let  $r_{i,b}^P$  and  $r_{i,b}^Q$  be the left side minus the right side of Eqs. (5) and (6). The equality residual is

$$\mathcal{L}_{\text{powerflow}} = \text{mean}_{b,i} [(r_{i,b}^P)^2 + (r_{i,b}^Q)^2]. \quad (25)$$

### Topology-conditioned graph Fourier neural operator



A line outage changes status, Laplacian, and admittance operators; learned parameters are shared.

Figure 1: Topology-conditioned GFNO-PINO architecture. A line outage changes the edge status, the graph Laplacian, and the exact admittance operators supplied to the loss; learned parameters are unchanged.

The two-ended thermal penalty is

$$\mathcal{L}_{\text{thermal}} = \text{mean}_{b,\ell} \left[ \text{ReLU} \left( \max \{ |\widehat{S}_{\ell,b}^f|, |\widehat{S}_{\ell,b}^t| \} - S_{\ell}^{\max} \right)^2 \right], \quad (26)$$

where inactive and unrated branches are masked. Although voltage boxes are enforced by the output layer, a diagnostic penalty is retained:

$$\mathcal{L}_{\text{voltage}} = \text{mean}_{b,i} \left[ \left( \text{ReLU}(V_i^{\min} - \widehat{V}_{i,b}) + \text{ReLU}(\widehat{V}_{i,b} - V_i^{\max}) \right)^2 \right]. \quad (27)$$

Given solver labels  $u_b^*$ , the warm-start term is

$$\mathcal{L}_{\text{data}} = \frac{1}{4} \sum_{q \in \{V, \theta, P^g, Q^g\}} \text{mean}_b \|\widehat{q}_b - q_b^*\|_2^2. \quad (28)$$

The total training objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cost}} \mathcal{L}_{\text{economic}} + \lambda_{\text{pf}} \mathcal{L}_{\text{powerflow}} + \lambda_{\text{line}} \mathcal{L}_{\text{thermal}} + \lambda_{\text{volt}} \mathcal{L}_{\text{voltage}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}. \quad (29)$$

The  $\lambda$  values are positive learned weights normalized to sum to the number of active objectives. Following gradient-norm balancing [29], let  $g_m = \|\nabla_{\Theta_s}(\lambda_m \mathcal{L}_m)\|_2$  for a shared backbone parameter  $\Theta_s$ , and let  $\rho_m = (\mathcal{L}_m / \mathcal{L}_{m,0}) / \text{mean}_j(\mathcal{L}_j / \mathcal{L}_{j,0})$ . The weight update minimizes

$$\mathcal{L}_{\text{GN}} = \sum_m |g_m - \bar{g} \rho_m^\alpha|, \quad \bar{g} = \text{mean}_m g_m, \quad (30)$$

while the model update treats the current normalized weights as constants. Every raw component and adaptive weight is logged separately.

### 3.5. Hard-constraint enforcement layer

For unconstrained decoder values  $a_i^V, a_g^P, a_g^Q \in \mathbb{R}$ , the independent box constraints are enforced by

$$\widehat{V}_i = V_i^{\min} + (V_i^{\max} - V_i^{\min}) \sigma(a_i^V), \quad (31)$$

$$\widehat{P}_g^g = P_g^{\min} + (P_g^{\max} - P_g^{\min}) \sigma(a_g^P), \quad (32)$$

$$\widehat{Q}_g^g = Q_g^{\min} + (Q_g^{\max} - Q_g^{\min}) \sigma(a_g^Q), \quad (33)$$

where  $\sigma$  is the logistic sigmoid. If  $\widetilde{\theta}$  is the raw angle vector, the reference is imposed by

$$\widehat{\theta}_i = \widetilde{\theta}_i - \widetilde{\theta}_{i_{\text{ref}}}. \quad (34)$$

Equations (31)–(34) guarantee the voltage and generator boxes and slack reference up to floating-point precision. They do not guarantee Eqs. (5), (6), or (13); those coupled non-convex constraints remain approximate and must be measured after inference.

### 3.6. Training procedure

Training uses two phases. The first phase supplies a stable solver-informed initialization. The second phase jointly optimizes economic quality and physical consistency, with adaptive weights reducing sensitivity to disparate units and gradient scales. Topology IDs are sampled only from the training partition.

## 4. Case study and experimental setup

### 4.1. Test systems

Experiments are configured for the IEEE 30-, 57-, 118-, and 300-bus systems distributed through pandapower and compatible MATPOWER data [23, 24]. Table 1 reports structural facts extracted by the released data pipeline.

Table 1: Test-system characteristics used by the data generator.

| Case     | Buses | Gen. | Loads | Lines | Trafos | Eligible N-1 |
|----------|-------|------|-------|-------|--------|--------------|
| IEEE 30  | 30    | 6    | 20    | 41    | 0      | 38           |
| IEEE 57  | 57    | 7    | 42    | 63    | 17     | 62           |
| IEEE 118 | 118   | 54   | 99    | 173   | 13     | 166          |
| IEEE 300 | 300   | 69   | 193   | 283   | 128    | 253          |

Table 2: Executed pilot training configurations. Width is the latent channel count and  $K$  is the Chebyshev order.

| Case     | Width | Depth | $K$ | Batch | Warm/physics epochs |
|----------|-------|-------|-----|-------|---------------------|
| IEEE 30  | 64    | 4     | 3   | 16    | 12/18               |
| IEEE 57  | 64    | 4     | 3   | 12    | 12/18               |
| IEEE 118 | 96    | 4     | 3   | 8     | 12/18               |
| IEEE 300 | 96    | 4     | 3   | 4     | 12/18               |

“Eligible N-1” counts non-islanding outages of elements represented as lines in each pandapower case; transformer outages are not included in the present contingency set.

All pilot runs use seed 2026, single-precision PyTorch 2.13.0 with CUDA 12.6, and an NVIDIA GeForce RTX 3050 6 GB Laptop GPU. Hyperparameters were fixed before inspecting test metrics. The checkpoint, configuration, parameter count, training history, software version, device, and runtime are archived for every run.

#### 4.2. Contingency and scenario generation

For every in-service line, the generator creates a copied network, sets that line out of service, and tests graph connectivity while retaining transformers and parallel circuits. Islanding outages are recorded and excluded. To maintain a fixed edge axis, topology artifacts preserve the base-case canonical branch list and set the outaged branch status to zero. Exact  $Y$ ,  $Y_f$ , and  $Y_t$  operators are stored once per topology; scenario files reference them by topology ID.

Loads are perturbed by bounded, spatially correlated multipliers. A squared-exponential kernel is formed from all-pairs graph distances and projected onto a positive-semidefinite eigenspectrum before sampling. Configured ranges are  $\pm 30\%$ ,  $\pm 35\%$ ,  $\pm 40\%$ , and  $\pm 20\%$  for the four pilot systems, respectively. The IEEE 300-bus range was reduced from the initially configured  $\pm 50\%$  after repeated IPOPT iteration-limit failures and is therefore a less stressed pilot, not a like-for-like stress test. Active and reactive demand at a load are scaled together to retain its base power factor. Renewable injections are allocated at generator or load buses and treated as fixed, unity-power-factor injections. All reported experiments use the generator’s correlated synthetic-diurnal profile, with capacity fractions 0.20, 0.20, 0.25, and 0.15, respectively. No claim of validation on real NREL or operational renewable data is made.

Each feasible scenario is solved with PowerModels’ polar AC formulation and IPOPT [5, 25]. Solver termination status, objective, tolerances, and wall-clock time are

retained. The pandapower/PYPOWER backend is available for pipeline debugging but must not be merged into a table labelled “IPOPT.” DC-OPF is solved separately for its speed and approximation baseline.

Topology IDs, not individual rows, are shuffled with a fixed seed and assigned 70%/15%/15% to train/validation/test. The base topology remains in training. Thus, every reported test contingency is absent from training. The intended pilot target is 12 converged scenarios for each of 12 selected topologies. The generator resamples failed scenarios up to a declared per-topology limit. The resulting generated, converged, and failed-attempt counts are reported in Section 5. The pilot is deliberately smaller than the planned full-contingency, multi-seed publication experiment.

#### 4.3. Baselines

The comparison includes:

1. IPOPT AC-OPF: ground-truth objective, constraint, and wall-clock reference.
2. DC-OPF: linear speed baseline; reactive and voltage feasibility are not attributed to this model.
3. Fixed-topology PINN: a dense PINN trained only on base-topology scenarios and evaluated without retraining on held-out outages.
4. Plain GNN: the same topology inputs and hard output head, with spatial mean aggregation replacing Chebyshev spectral filters.
5. Data-only GFNO: the proposed graph-spectral architecture trained throughout on solver-label mean-square error, without physics fine-tuning.

The fixed-topology PINN establishes the cost of omitting topology; the plain GNN isolates the spectral component; and the data-only GFNO isolates the PINO loss. Model width and depth are shared where the architecture permits it. Parameter counts are retained with each checkpoint because the dense fixed PINN and spatial GNN are not exactly parameter-matched to the GFNO.

---

**Algorithm 1** Supervised warm start and physics-informed fine-tuning

---

Require: Topology-disjoint datasets  $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}$ ; model  $\mathcal{G}_{\Theta}$ ; warm epochs  $E_w$ ; physics epochs  $E_p$

- 1: Initialize  $\Theta$ , optimizer, scheduler, and adaptive log-weights  $\{\log \lambda_m\}$
- 2: for  $e = 1$  to  $E_w$  do
- 3:   for mini-batch  $\mathcal{B} \subset \mathcal{D}_{\text{tr}}$  do
- 4:     Predict  $\hat{u}_b = \mathcal{G}_{\Theta}(X_b, E_b, z_b, \mathcal{M}_b)$
- 5:     Update  $\Theta$  using  $\mathcal{L}_{\text{data}}$
- 6:   end for
- 7:   Evaluate label error and AC residuals on  $\mathcal{D}_{\text{val}}$
- 8: end for
- 9: Record initial physics losses  $\{\mathcal{L}_{m,0}\}$
- 10: for  $e = 1$  to  $E_p$  do
- 11:   for mini-batch  $\mathcal{B} \subset \mathcal{D}_{\text{tr}}$  do
- 12:     Predict  $\hat{u}_b$  with sample-specific topology
- 13:     Compute Eqs. (24)–(28) using  $Y_b, Y_{f,b}, Y_{t,b}$
- 14:     Update  $\{\log \lambda_m\}$  using Eq. (30)
- 15:     Update  $\Theta$  using Eq. (29); clip the global gradient norm
- 16:   end for
- 17:   Save the checkpoint minimizing validation data error plus power-flow residual
- 18: end for
- 19: return Best validation checkpoint and all component-wise logs

---

#### 4.4. Evaluation metrics

For sample  $b$ , the signed optimality gap is

$$\Delta C_b(\%) = 100 \frac{C(\hat{P}_b^g) - C(P_b^{g,*})}{\max(|C(P_b^{g,*})|, \epsilon)}. \quad (35)$$

A negative value does not establish superiority if the prediction is infeasible. Active and reactive KCL residuals are reported per bus, with sample mean and maximum. KVL cycle sums are reported as a numerical diagnostic; they are structurally zero because the model outputs bus angle potentials.

For constraint class  $c$ , the violation rate and magnitude are

$$R_c = \frac{\#\{k : v_{c,k} > \tau_c\}}{\#\{k\}}, \quad (36)$$

$$M_c^{\max} = \max_k v_{c,k}, \quad M_c^{\text{mean}} = \text{mean}_k v_{c,k}, \quad (37)$$

for voltage, thermal, and generator constraints. The numerical threshold is  $10^{-6}$  p.u. for these inequalities and  $10^{-3}$  p.u. for KCL. Latency is measured after ten warm-up passes with batch size one and CUDA device synchronization, then summarized across the 36 test samples. Results are grouped by bus count and topology ID. Accuracy on the test partition is the primary unseen-topology result; random row-wise splits are not reported as contingency generalization.

## 5. Results and discussion

### 5.1. Executed pilot and numerical verification

The complete pilot pipeline produced 564 converged AC-OPF labels and 576 neural test evaluations. All AC references were solved with PowerModels 0.21.6, JuMP 1.31.1, and IPOPT 1.15.0 using the polar formulation, an optimality tolerance and constraint-violation tolerance of  $10^{-8}$ , and the default MUMPS linear solver. The model code passed 15 automated tests, including an analytic two-bus balance, independent explicit  $G$ – $B$  injection equations, two-ended branch flows, automatic differentiation, hard-box enforcement, topology-disjoint splitting, outage masking on a canonical edge axis, the MATPOWER writer, and comparison with a converged IEEE 30-bus power flow.

Table 3 exposes two important selection effects. First, the IEEE 30 outage line\_9\_out produced no converged sample in 60 attempts and was retained only as a zero-count training topology; it did not enter the test set. Second, the imported IEEE 57 and IEEE 300 cases required explicit voltage ranges of  $[0.8, 1.1]$  and  $[0.9, 1.1]$  p.u., respectively, to obtain a runnable pilot. Together with the reduced IEEE 300 load range, these adjustments mean that cross-system comparisons measure computational scaling, not identical electrical stress. Every test set nevertheless contains three contingencies absent from its corresponding training set.

### 5.2. Economic performance

Table 4 reports the signed cost gap defined in Eq. (35). The GFNO-PINO gaps are  $-20.71\%$ ,  $-2.93\%$ ,  $-29.46\%$ , and  $-22.85\%$  from 30 to 300 buses, whereas its mean absolute gaps are 21.44%, 13.28%, 30.01%, and 22.85%. These negative signed gaps are caused by infeasible predictions with artificially low generation cost; they are not evidence that the network has a better feasible operating point than IPOPT. The same caution applies to the DC-OPF gaps of  $-2.65\%$  to  $-2.03\%$ , because DC-OPF omits AC losses, voltage, and reactive constraints. Since the neural AC-feasibility rate is zero, the feasibility-conditioned optimality gap is undefined for every neural method. Consequently, the pilot does not establish economic superiority.

### 5.3. Physical residuals and constraint violations

The hard output transformations behave as intended: voltage and generator box-violation rates are zero for every neural architecture and system. This is a parameterization check, not learned feasibility. GFNO-PINO also has zero measured thermal excess on the pilot test sets. Its mean sample-maximum KCL residuals remain 0.228, 1.307, 1.250, and 3.232 p.u.; the corresponding worst-sample values are 0.267, 2.810, 1.254, and 3.845 p.u. Using a maximum-KCL threshold of  $10^{-3}$  p.u. together with zero box, generator, and thermal excess, no GFNO-PINO prediction satisfies the AC-feasibility proxy.

Physics fine-tuning nevertheless provides a strong residual effect. Relative to the data-only GFNO, it reduces

Table 3: Executed topology-disjoint pilot dataset. “Failed” counts solver attempts rejected before the target number of converged labels was reached.

| Case     | Target | Converged | Failed | Usable topol. | Train | Validation | Test | Test topol. |
|----------|--------|-----------|--------|---------------|-------|------------|------|-------------|
| IEEE 30  | 144    | 132       | 154    | 11            | 84    | 12         | 36   | 3           |
| IEEE 57  | 144    | 144       | 46     | 12            | 96    | 12         | 36   | 3           |
| IEEE 118 | 144    | 144       | 0      | 12            | 96    | 12         | 36   | 3           |
| IEEE 300 | 144    | 144       | 0      | 12            | 96    | 12         | 36   | 3           |

Table 4: Pilot signed optimality gap (%) relative to IPOPT on topology-disjoint test sets. Negative neural gaps are infeasible lower-cost predictions, not superior optima.

| Method         | IEEE 30 | IEEE 57 | IEEE 118 | IEEE 300 |
|----------------|---------|---------|----------|----------|
| DC-OPF         | -2.65   | -3.16   | -2.94    | -2.03    |
| Fixed PINN     | 4.42    | -2.51   | 25.96    | 0.15     |
| Plain GNN      | -12.64  | -4.94   | -23.49   | -17.74   |
| Data-only GFNO | 2.22    | 1.96    | 31.46    | 1.54     |
| GFNO-PINO      | -20.71  | -2.93   | -29.46   | -22.85   |

mean sample-maximum KCL by 79.0%, 47.0%, 79.6%, and 96.4% for the four systems. The 300-bus comparison is especially large (90.521 to 3.232 p.u.), showing that label matching alone can conceal a severe electrical inconsistency. Because bus angles are scalar potentials, cycle KVL is structural in all models and is not counted as a learned improvement. The remaining KCL mismatch demonstrates why box-constrained heads and small supervised errors cannot substitute for an AC feasibility layer.

Figure 2 shows that the curriculum transition changes the validation-residual trajectory rather than merely extending supervised fitting. Figure 3 also shows that the adaptive coefficients remain positive and finite; at the final epoch the raw power-flow weights range from 0.283 to 0.625 rather than collapsing to zero. These curves support the residual-reduction observation, but their non-zero end values do not imply convergence to the AC equality manifold.

#### 5.4. Inference latency and scalability

The workstation contains a 13th-generation Intel Core i5-13420H CPU and an NVIDIA GeForce RTX 3050 6 GB Laptop GPU. IPOPT and DC-OPF timings are CPU solver times stored during data generation; neural timings use synchronized CUDA inference in float32 after ten warm-up passes. Medians and interquartile ranges are calculated across the 36 unseen-topology samples per system. Thus, the comparison describes end-to-end alternatives on one workstation, not identical CPU-only hardware.

The median GFNO-PINO/IPOPT ratios correspond to 15.6, 19.0, 119.9, and 131.2 times lower latency from 30 to 300 buses. GFNO-PINO is also faster than the stored DC-OPF solve in every case. Its own median latency increases from 3.66 to 11.31 ms, with non-monotonic intermediate values caused by small test sets, GPU launch overhead, and dense padded graph operations. These timing gains

cannot be claimed as operational AC-OPF speedups because all neural predictions fail the AC-feasibility proxy. Any feasibility-restoration or verification solve must be included in a deployment latency.

#### 5.5. Unseen-topology generalization and ablation

Table 7 averages the four system-level test results. The physics loss is the clearest successful ablation: the data-only GFNO has a four-case mean sample-maximum KCL residual of 25.049 p.u., compared with 1.504 p.u. for GFNO-PINO. The spatial GNN, however, averages 1.529 p.u. and is better than GFNO-PINO on IEEE 57 (1.189 versus 1.307 p.u.) and IEEE 118 (1.056 versus 1.250 p.u.). GFNO-PINO is better on IEEE 30 (0.228 versus 0.278 p.u.) and IEEE 300 (3.232 versus 3.591 p.u.). The pilot therefore supports topology conditioning and physics-informed training, but does not establish a consistent advantage for Chebyshev spectral filtering over spatial message passing.

The fixed PINN has the highest four-case average thermal violation rate (3.7%) and a mean maximum-KCL residual of 2.476 p.u. Its residual is worse than both topology-aware physics models, consistent with distribution shift under unseen admittances. Nevertheless, all results come from one seed and one checkpoint per case. Training variance, topology difficulty, and unequal parameter counts are confounded, so architectural significance cannot be claimed without repeated seeds and a larger held-out contingency set.



Table 5: GFNO-PINO pilot constraint diagnostics on unseen topologies. Box and thermal cells report violation rate (%) / maximum magnitude; AC feasible uses a  $10^{-3}$  p.u. maximum-KCL threshold.

| System   | Voltage    | Thermal    | Generator  | Mean max. KCL (p.u.) | AC feasible (%) |
|----------|------------|------------|------------|----------------------|-----------------|
| IEEE 30  | 0.00/0.000 | 0.00/0.000 | 0.00/0.000 | 0.228                | 0.0             |
| IEEE 57  | 0.00/0.000 | 0.00/0.000 | 0.00/0.000 | 1.307                | 0.0             |
| IEEE 118 | 0.00/0.000 | 0.00/0.000 | 0.00/0.000 | 1.250                | 0.0             |
| IEEE 300 | 0.00/0.000 | 0.00/0.000 | 0.00/0.000 | 3.232                | 0.0             |

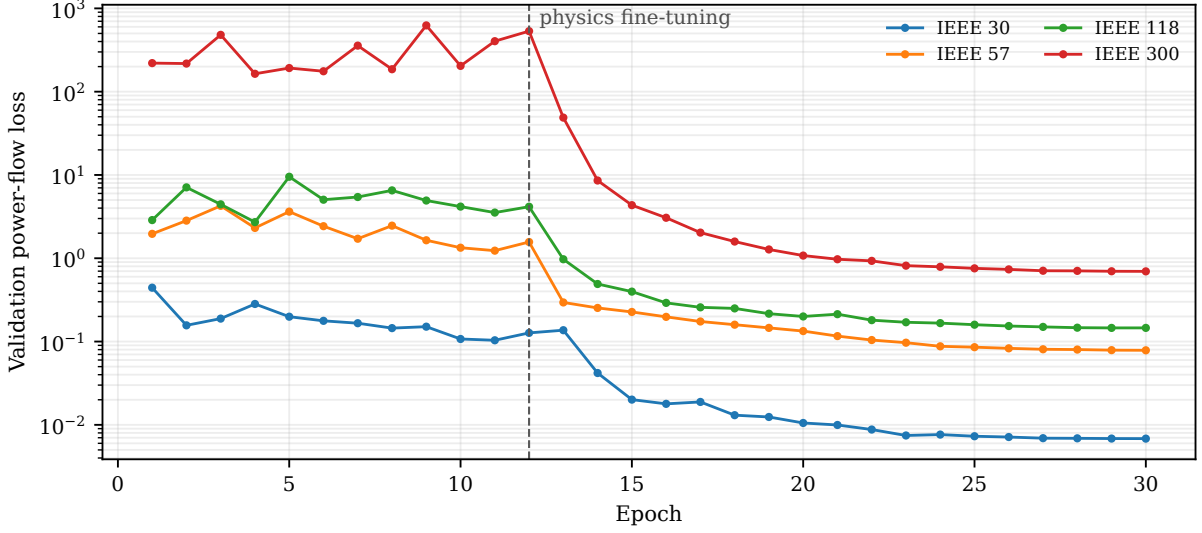


Figure 2: Validation power-flow residual during the 12-epoch supervised warm start and 18-epoch physics fine-tuning. Each curve uses topology-disjoint validation samples for one IEEE system.

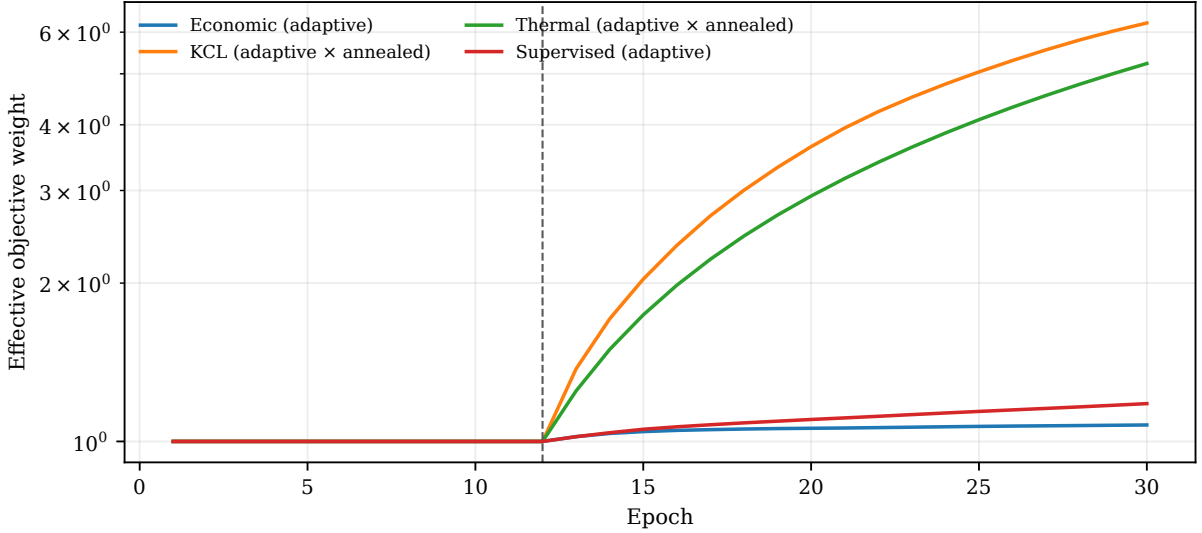


Figure 3: Illustrative adaptive loss weights for the IEEE 30 GFNO-PINO run. The power-flow and thermal curriculum multipliers are applied in addition to the learned positive weights during physics fine-tuning; the other run histories are archived separately.

Table 6: Pilot batch-one wall-clock latency in milliseconds: median [interquartile range] after warm-up on the same machine.

| System   | IPOPT AC-OPF               | DC-OPF                  | Fixed PINN        | Plain GNN         | GFNO-PINO            |
|----------|----------------------------|-------------------------|-------------------|-------------------|----------------------|
| IEEE 30  | 57.29 [51.60, 68.77]       | 25.52 [24.54, 31.35]    | 1.51 [0.91, 1.68] | 3.16 [2.96, 4.01] | 3.66 [3.32, 4.32]    |
| IEEE 57  | 204.45 [193.95, 225.83]    | 123.24 [112.48, 151.16] | 1.71 [0.71, 1.84] | 8.12 [5.93, 8.81] | 10.75 [6.14, 13.00]  |
| IEEE 118 | 524.17 [471.33, 601.95]    | 155.79 [143.19, 170.88] | 1.74 [1.63, 1.87] | 2.10 [2.01, 2.58] | 4.37 [3.47, 5.77]    |
| IEEE 300 | 1484.22 [1384.81, 1634.85] | 191.22 [173.52, 205.12] | 0.56 [0.54, 0.59] | 2.43 [2.29, 2.68] | 11.31 [10.81, 12.67] |

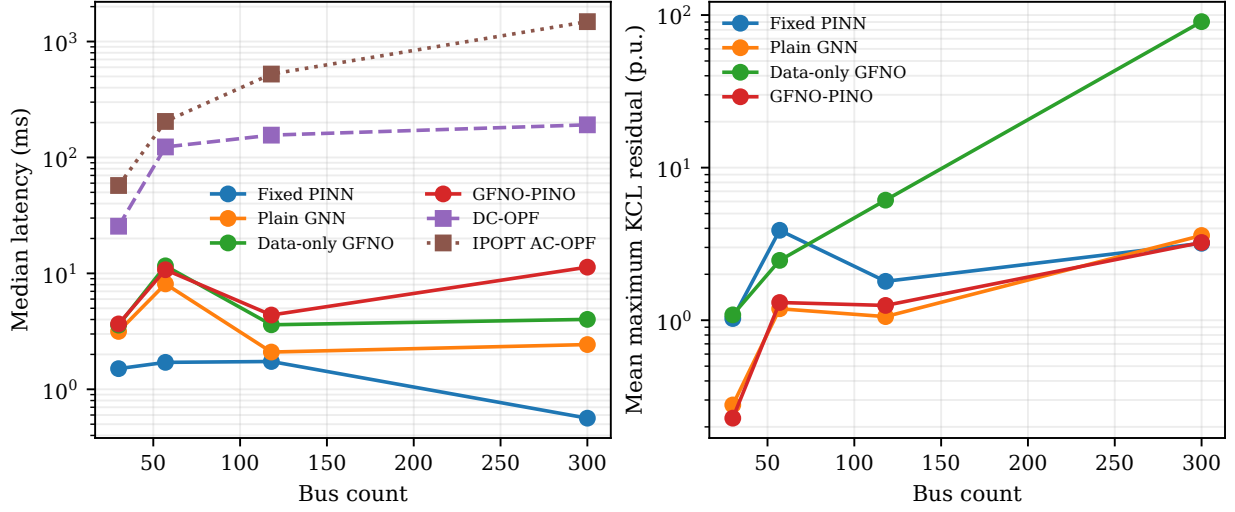


Figure 4: Pilot scaling of median batch-one latency and mean sample-maximum KCL residual. Error bars show interquartile latency ranges or the dispersion across test samples, not independent training seeds.

Table 7: Four-case pilot ablation averages. Result is mean absolute gap (%) / mean maximum KCL residual (p.u.) / thermal violation rate (%).

| Model          | Topology input | Spectral layer | Physics loss | Hard boxes | Result          |
|----------------|----------------|----------------|--------------|------------|-----------------|
| Fixed PINN     | No             | No             | Yes          | Yes        | 14.1/2.476/3.7  |
| Plain GNN      | Yes            | No             | Yes          | Yes        | 17.7/1.529/0.0  |
| Data-only GFNO | Yes            | Yes            | No           | Yes        | 15.0/25.049/1.4 |
| GFNO-PINO      | Yes            | Yes            | Yes          | Yes        | 21.9/1.504/0.0  |

## 6. Limitations

The proposed output layer guarantees independent voltage and generator bounds and the reference angle, but it does not project onto the full non-convex AC feasible set. KCL is optimized as a soft objective and remains far above the declared tolerance; any deployment requires explicit verification and an AC correction solve. The zero thermal excess observed for GFNO-PINO does not repair equality infeasibility. The pilot uses only 12 selected topologies and one random seed per system, synthetic renewable profiles, and 30 total epochs. It is therefore an implementation and trend study rather than the final full-contingency statistical benchmark.

The method is evaluated only through 300 buses, and the current dense Laplacian batching path is not evidence of scalability to substantially larger interconnections. Although the parameterization is node- and edge-based, the experiments use one checkpoint per grid size; they do not demonstrate one shared checkpoint across 30–300 buses. Topology independence is limited to non-islanding single-line outages within compatible network families and does not cover bus additions, generator-set changes, islands, or simultaneous N-k contingencies. The relaxed IEEE 57/300 voltage ranges and reduced IEEE 300 load range further limit direct cross-case stress comparisons. Finally,

IPOPT provides locally converged references rather than global-optimality certificates, and the rejected scenarios—particularly the 154 IEEE 30 failures—may introduce selection bias.

## 7. Conclusion

This paper formulates real-time AC-OPF as a topology-conditioned operator learning problem and develops a physics-informed graph Fourier neural operator for that setting. The central design choice is a Chebyshev polynomial spectral filter of the sample-specific normalized Laplacian. It preserves a graph spectral inductive bias without requiring eigenvector alignment after line outages. Explicit branch status and admittance features make contingencies operator inputs, while exact  $Y$ ,  $Y_f$ , and  $Y_l$  matrices define differentiable AC residuals. Bounded output transformations enforce voltage and generator boxes and the slack reference; they are deliberately not presented as a complete AC-feasibility guarantee.

The executed pilot used 564 IPOPT labels and 144 topology-disjoint neural test scenarios across four IEEE systems. Physics fine-tuning reduced mean sample-maximum KCL by 47.0–96.4% relative to the data-only GFNO, eliminated observed thermal excess for GFNO-PINO, and retained millisecond inference. However, every neural architecture had a 0% AC-feasibility rate at the  $10^{-3}$  p.u. maximum-KCL threshold. Cost gaps therefore cannot be interpreted as feasible optimality results. The Chebyshev GFNO also did not consistently beat the spatial GNN. The defensible conclusion is that topology-conditioned PINO is a promising residual-reducing warm-start or screening component, not a stand-alone AC-OPF solver in its present form.

The next step is an explicit differentiable feasibility-restoration layer or post-network Newton/AC-OPF correction, followed by full N-1 coverage, at least five seeds, real renewable time series, sparse mixed-size batching, and hardware-matched latency accounting. N-k, island-aware operation, uncertainty quantification, and validation on operational networks remain open. Until those steps produce verified feasible outputs, independent solver validation is necessary for safety-critical dispatch.

## Data and code availability

The accompanying workspace contains the implementation, configuration files, data-generation scripts, exact pilot configurations, solver manifests, checkpoints, per-sample evaluations, aggregate CSV files, and figure-generation routine. These artifacts are supplied with the review package. Any public archive must preserve the licenses of the source IEEE/MATPOWER cases.

## References

- [1] S. Frank, S. Rebennack, An introduction to optimal power flow: Theory, formulation, and examples, *IIE Transactions* 48 (12) (2016) 1172–1197. doi:10.1080/0740817X.2016.1189626.
- [2] F. Capitanescu, Critical review of recent advances and further developments needed in AC optimal power flow, *Electric Power Systems Research* 136 (2016) 57–68. doi:10.1016/j.epsr.2016.02.008.
- [3] D. K. Molzahn, I. A. Hiskens, A survey of relaxations and approximations of the power flow equations, *Foundations and Trends in Electric Energy Systems* 4 (1–2) (2019) 1–221. doi:10.1561/31000000012.
- [4] F. Zohrizadeh, C. Josz, M. Jin, R. Madani, J. Lavaei, S. Sojoudi, A survey on conic relaxations of optimal power flow problem, *European Journal of Operational Research* 287 (2) (2020) 391–409. doi:10.1016/j.ejor.2020.01.034.
- [5] A. Wächter, L. T. Biegler, On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming, *Mathematical Programming* 106 (1) (2006) 25–57. doi:10.1007/s10107-004-0559-y.
- [6] B. Stott, J. Jardim, O. Alsaç, DC power flow revisited, *IEEE Transactions on Power Systems* 24 (3) (2009) 1290–1300. doi:10.1109/TPWRS.2009.2021235.
- [7] X. Pan, T. Zhao, M. Chen, S. Zhang, DeepOPF: A deep neural network approach for security-constrained DC optimal power flow, *IEEE Transactions on Power Systems* 36 (3) (2021) 1725–1735. doi:10.1109/TPWRS.2020.3026379.
- [8] X. Pan, M. Chen, T. Zhao, S. H. Low, DeepOPF: A feasibility-optimized deep neural network approach for AC optimal power flow problems, *IEEE Systems Journal* 17 (1) (2023) 673–683. doi:10.1109/JSYST.2022.3201041.
- [9] M. Zhou, M. Chen, S. H. Low, DeepOPF-FT: One deep neural network for multiple AC-OPF problems with flexible topology, *IEEE Transactions on Power Systems* 38 (1) (2023) 964–967. doi:10.1109/TPWRS.2022.3217407.
- [10] F. Fioretto, T. W. K. Mak, P. Van Hentenryck, Predicting AC optimal power flows: Combining deep learning and lagrangian dual methods, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34, 2020, pp. 630–637. doi:10.1609/aaai.v34i01.5403.
- [11] A. Venzke, G. Qu, S. H. Low, S. Chatzivasileiadis, Learning optimal power flow: Worst-case guarantees for neural networks, in: *2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, 2020, pp. 1–7. doi:10.1109/SmartGridComm47815.2020.9302963.
- [12] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *Journal of Computational Physics* 378 (2019) 686–707. doi:10.1016/j.jcp.2018.10.045.
- [13] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang, Physics-informed machine learning, *Nature Reviews Physics* 3 (6) (2021) 422–440. doi:10.1038/s42254-021-00314-5.
- [14] G. S. Misyris, A. Venzke, S. Chatzivasileiadis, Physics-informed neural networks for power systems, in: *2020 IEEE Power & Energy Society General Meeting*, 2020, pp. 1–5. doi:10.1109/PESGM41954.2020.9282004.
- [15] R. Nellikkath, S. Chatzivasileiadis, Physics-informed neural networks for AC optimal power flow, *Electric Power Systems Research* 212 (2022) 108412. doi:10.1016/j.epsr.2022.108412.
- [16] X. Lei, Z. Yang, J. Yu, J. Zhao, Q. Gao, H. Yu, Data-driven optimal power flow: A physics-informed machine learning approach, *IEEE Transactions on Power Systems* 36 (1) (2021) 346–354. doi:10.1109/TPWRS.2020.3001919.
- [17] L. Lu, P. Jin, G. Pang, Z. Zhang, G. E. Karniadakis, Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators, *Nature Machine Intelligence* 3 (3) (2021) 218–229. doi:10.1038/s42256-021-00302-5.
- [18] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar, Fourier neural operator for parametric partial differential equations, in: *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=c8P9NQVtmnO>
- [19] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, A. Anandkumar, Neural operator: Learning maps between function spaces with applications to PDEs, *Journal of Machine Learning Research* 24 (89) (2023) 1–97. URL <https://jmlr.org/papers/v24/21-1524.html>
- [20] D. K. Hammond, P. Vandergheynst, R. Gribonval, Wavelets on graphs via spectral graph theory, *Applied and Computational Harmonic Analysis* 30 (2) (2011) 129–150. doi:10.1016/j.acha.2010.04.005.
- [21] M. Defferrard, X. Bresson, P. Vandergheynst, Convolutional neural networks on graphs with fast localized spectral filtering, in: *Advances in Neural Information Processing Systems*, Vol. 29, 2016, pp. 3837–3845. URL <https://proceedings.neurips.cc/paper/2016/hash/04df4d434d481c5bb723be1b6df1ee65-Abstract.html>
- [22] M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, P. Vandergheynst, Geometric deep learning: Going beyond euclidean data, *IEEE Signal Processing Magazine* 34 (4) (2017) 18–42. doi:10.1109/MSP.2017.2693418.
- [23] R. D. Zimmerman, C. E. Murillo-Sánchez, R. J. Thomas, MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education, *IEEE Transactions on Power Systems* 26 (1) (2011) 12–19. doi:10.1109/TPWRS.2010.2051168.
- [24] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dolichon, F. Meier, S. Meinecke, M. Braun, Pandapower—an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems, *IEEE Transactions on Power Systems* 33 (6) (2018) 6510–6521. doi:10.1109/TPWRS.2018.2829021.
- [25] C. Coffrin, R. Bent, K. Sundar, Y. Ng, M. Lubin, PowerModels.jl: An open-source framework for exploring power flow formulations, in: *2018 Power Systems Computation Conference (PSCC)*, 2018, pp. 1–8. doi:10.23919/PSCC.2018.8442948.
- [26] T. N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: *International Conference on Learning Representations*, 2017. URL <https://openreview.net/forum?id=SJU4ayYgl>
- [27] B. Donon, B. Donnot, I. Guyon, A. Marot, Graph neural solver for power systems, in: *2019 International Joint Conference on Neural Networks (IJCNN)*, 2019, pp. 1–8. doi:10.1109/IJCNN.2019.8851855.
- [28] D. Owerko, F. Gama, A. Ribeiro, Optimal power flow using graph neural networks, in: *ICASSP 2020–2020 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2020, pp. 5930–5934. doi:10.1109/ICASSP40776.2020.9053140.
- [29] Z. Chen, V. Badrinarayanan, C.-Y. Lee, A. Rabinovich, GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks, in: *Proceedings of the 35th International Conference on Machine Learning*, Vol. 80 of *Proceedings of Machine Learning Research*, 2018, pp. 794–803. URL <https://proceedings.mlr.press/v80/chen18a.html>